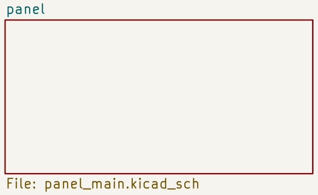


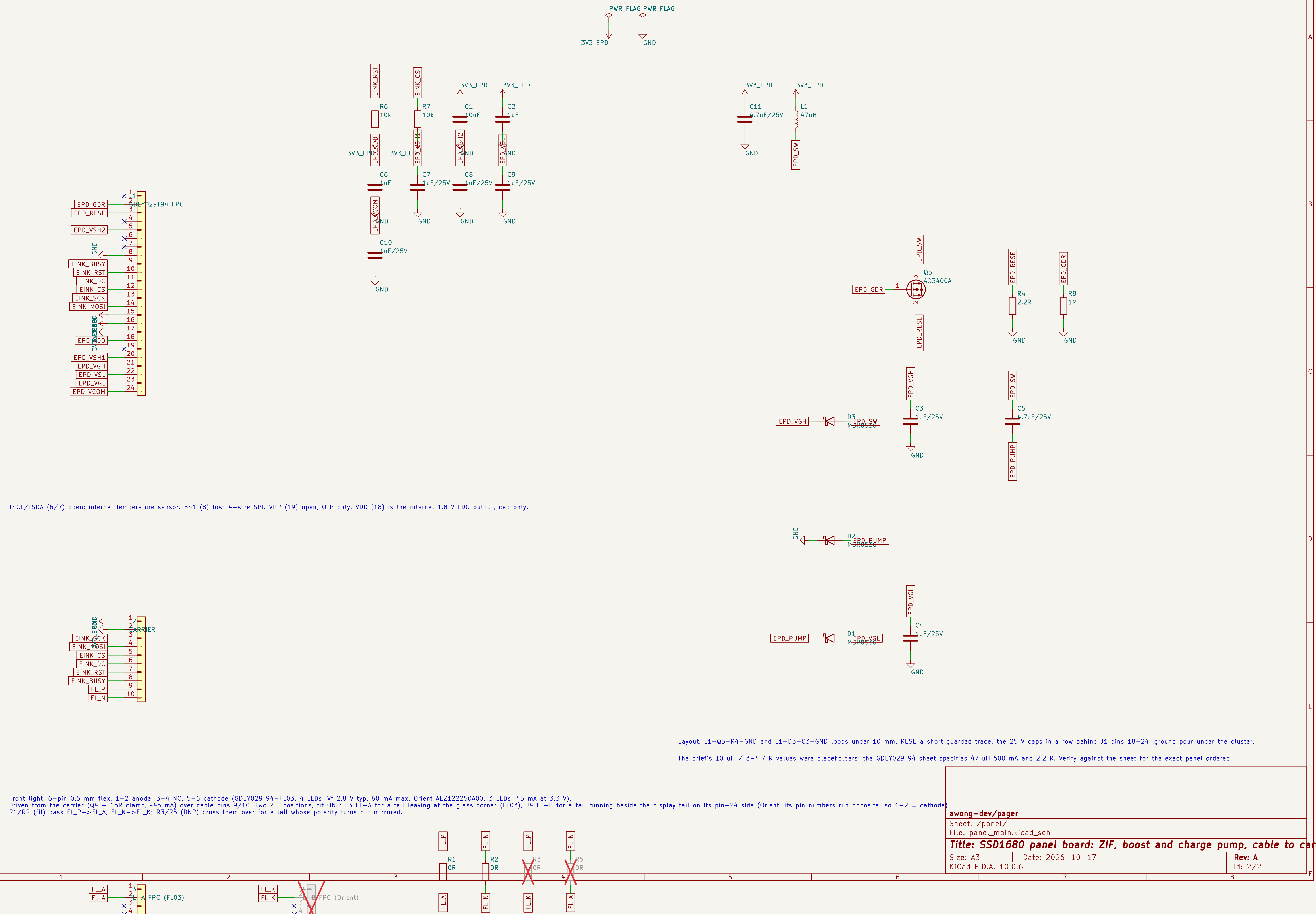
awong-dev/pager
Sheet: /
File: pager-panel.kicad_sch

Title: pager-panel

Size: A4
Date: 2026-10-17
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/2

Panel board for 24-pin 0.5 mm SSD1680/SSD1681 e-paper (Good Display GDEY029T94 2.9" 296x128, Orient AEZ122250A00 2.13" 122x250, ...). Reference circuit identical on both sheets: RESE 2.2R, L 47uH, MBR0530, 25 V caps, 4-wire SPI.

Boost: 3V3_EPD -> L1 -> SW node, Q5 (gate GDR) switches SW to GND through R4 (RESE sense). D3 SW->VGH into C3. C5 flies SW->PUMP; D2 clamps PUMP to GND (a



TSCL/TSDA (6/7) open: internal temperature sensor. BS1 (8) low: 4-wire SPI. VPP (19) open, OTP only. VDD (18) is the internal 1.8 V LDO output, cap only.

Layout: L1-Q5-R4-GND and L1-D3-C3-GND loops under 10 mm; RESE a short guarded trace; the 25 V caps in a row behind J1 pins 18-24; ground pour under the cluster.

The brief's 10 uH / 3-4.7 R values were placeholders; the GDEY029T94 sheet specifies 47 uH 500 mA and 2.2 R. Verify against the sheet for the exact panel ordered.

Front light: 6-pin 0.5 mm flex, 1-2 anode, 3-4 NC, 5-6 cathode (GDEY029T94-FL03: 4 LEDs, Vf 2.8 V typ, 60 mA max; Orient AEZ122250A00: 3 LEDs, 45 mA at 3.3 V).
Driven from the carrier (Q4 + 15R clamp, ~45 mA) over cable pins 9/10. Two ZIF positions, fit ONE: J3 FL-A for a tail leaving at the glass corner (FL03), J4 FL-B for a tail running beside the display tail on its pin-24 side (Orient; its pin numbers run opposite, so 1-2 = cathode).
R1/R2 (fit) pass FL_P->FL_A, FL_N->FL_K; R3/R5 (DNP) cross them over for a tail whose polarity turns out mirrored.

awong-dev/pager
Sheet: /panel/
File: panel_main.kicad_sch
Title: SSD1680 panel board: ZIF, boost and charge pump, cable to carrier
Size: A3 Date: 2026-10-17 Rev: A
KiCad E.D.A. 10.0.6 Id: 2/2